Conservation and Specialization in Semaev-Based Index Calculus over Prime Fields

Anonymous

Abstract

Let E be an elliptic curve over a field K and let S_m be Semaev's m-th summation polynomial. We determine the Galois group of $S_m(t_1, \ldots, t_{m-1}, T)$ over $K(t_1, \ldots, t_{m-1})$ for every $m \geq 3$: it is elementary abelian of order 2^{m-2}, acting simply transitively on the 2^{m-2} roots, and it is the same group for every elliptic curve, every base field and every characteristic. The proof identifies the splitting field with the function field of $E^{m-1}/\langle -1 \rangle$, where -1 acts diagonally; the group is the deck group $(\mathbb{Z}/2)^{m-1}$ of $E^{m-1} \to (E/\pm)^{m-1}$ modulo the diagonal. Geometric and arithmetic monodromy coincide.

Two consequences are recorded. First, an exact factorization law over a finite field $\mathbb{F}_q$: at every specialization outside an explicitly described degenerate locus of size $O_m(q^{m-2})$, $S_m(a_1, \ldots, a_{m-1}, T)$ either splits into 2^{m-2} distinct linear factors — exactly when the $m-1$ fibres $\pi^{-1}(a_i)$ of the x-map are simultaneously rational or simultaneously irrational — or is a product of 2^{m-3} distinct irreducible quadratics. No irreducible factor of degree ≥ 3 ever occurs, and the factor degrees are always equal. Second, the factor-base locus, on which index-calculus relation search operates, lies entirely inside the totally split locus, at every m.

Combined with an elementary conservation identity — the total number of m-term decompositions over a factor base of fixed size is a binomial coefficient, so mean decomposition yield is independent of factor-base geometry — these give a coherent negative statement: generic-fibre arithmetic statistics and factor-base relation statistics are different objects, and several proposed relation-rate levers are fixed either by conservation or by complete splitting on the rational-point locus. Nothing here claims an ECDLP speedup, a security reduction, or a new hardness result; several of the ingredients are classical and are marked as such. All statements are verified computationally on 20 curves over 6 primes at $m = 3, 4, 5$, including 257 061 exhaustive specializations with zero discrepancies, against null controls that the same instrument rejects.

Contents

1 Introduction **2**

2 Notation and the summation cover **3**

3 The structure theorem **4**
3.1 The two lemmas 4
3.2 The theorem 5
3.3 Corollaries 6
3.4 The branch locus 7
3.5 Special j-invariants, complex multiplication, and characteristic 7

4 An exact factorization law over a finite field **8**

5 Specialization to the factor-base locus **9**
5.1 Why complete splitting is not an algorithm 10

6 Conservation of decomposition yield **10**

7 Synthesis: which levers are fixed and which are free **11**

8 Computational verification **12**

9 Relation to prior work, and the limits of this novelty assessment **13**

10 Open questions **14**

1 Introduction

Semaev's summation polynomials [7] translate the statement "m points of an elliptic curve sum to the identity" into a polynomial condition on their x-coordinates alone, and they are the algebraic engine of index-calculus attacks on the elliptic curve discrete logarithm problem (ECDLP) [5, 1, 2]. A recurring hope in that line of work is that some *choice* — of curve, of curve family, of factor base — might raise the rate at which relations are found. This paper makes three statements about which choices can possibly matter, and finds that a specific, identifiable class of them cannot.

The main object is the cover of parameter space cut out by S_m in its last variable. Write $n = m - 1$, let $t_1, \dots, t_n$ be independent transcendentals over K, and consider

$$S_m(t_1, \dots, t_n, T) \in K(t_1, \dots, t_n)[T],$$

a polynomial of degree 2^{m-2} in T. Theorem 3.4 says its Galois group is $(\mathbb{Z}/2)^{m-2}$ in its regular representation, for every E, every K and every characteristic, with geometric monodromy equal to arithmetic monodromy.

The proof is short once the right object is named. The roots of $S_m(t_1, \dots, t_n, T)$ are the x-coordinates of the signed sums $\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n$ of the tautological points P_i with $x(P_i) = t_i$ — a classical observation, immediate from the defining property of S_m. The splitting field is therefore contained in $K(E^n)$, which is Galois over $K((E/\pm)^n) = K(t_1, \dots, t_n)$ with group $(\mathbb{Z}/2)^n$, the deck group of $E^n \to (E/\pm)^n$; no computation with square classes is needed, and the argument is insensitive to the characteristic. The diagonal -1 fixes every root because $x(R) = x(-R)$, and — this is the only step with content — nothing else does, because a further coincidence among the roots would force a nontrivial relation $\sum_{i \in S} \varepsilon_i P_i \in E[2]$ at the generic point of E^n, which is a proper closed condition. So the group is $(\mathbb{Z}/2)^n$ modulo the diagonal. Because the group is abelian, the stabilizer of a root is normal, so the root field is already the splitting field: the cover is Galois.

Over a finite field this yields an exact factorization law (Theorem 4.2). Frobenius acts on the deck group through the $(m-1)$-tuple recording, for each i, whether the fibre $\pi^{-1}(a_i)$ of the x-map is $\mathbb{F}_q$-rational. Because the action of the group on the roots is regular, every nonidentity element is fixed-point-free, so the fibre is either totally split or has no rational point at all: the dichotomy in the abstract. This is a sharp, immediately checkable statement about a heavily studied family of polynomials.

Specializing to the locus where each a_i is the x-coordinate of a *rational* point — the factor-base locus — all the rationality invariants agree and the fibre splits completely, at every m (Theorem 5.1). This is the algebraic-geometry restatement of a fact index calculus already uses operationally, and we are careful in §5.1 to explain why it is not an algorithm. Its interest is negative and methodological: a Frobenius/Chebotarev census of the generic fibre measures a statistic that is *identically constant* on the locus where relation search actually operates, so such a census cannot detect a relation-rate anomaly there — there is none to detect.

Section 6 records the complementary elementary fact on the combinatorial side: a double count fixes the mean number of m-term decompositions per target at $\binom{B+m-1}{m}/N$ for *every* factor base of size B in a group of order N. Mean yield is not a design lever; only the distribution is free.

What this paper does not claim

It claims no improvement to any ECDLP algorithm, no security reduction, no new hardness assumption, and no change to any complexity estimate. Every asymptotic statement about index calculus in the literature stands untouched. The results are structural facts about a family of polynomials, plus one combinatorial identity, and their content for cryptanalysis is exclusively that of ruling certain proposals out.

It also claims no novelty for its classical ingredients. The identification of the roots as signed sums, the degree 2^{m-2}, the symmetry, the absolute irreducibility of S_m, the resultant recursion, and the operational "solve for the last variable" step are all standard [7, 5, 6]. What is offered as possibly new is the Galois-theoretic identification for all m (Theorem 3.4), the exact factorization law (Theorem 4.2), and the synthesis in §7. §9 states honestly what search was and was not possible from the environment in which this work was done, and does not assert novelty beyond what that search supports.

2 Notation and the summation cover

Throughout, K is a field and E/K is an elliptic curve with origin O and inversion $\iota = [-1]$. Let

$$\pi\colon E \longrightarrow E/\langle\iota\rangle \cong \mathbb{P}^1_K$$

be the quotient map. We use only the following two properties of π, both of which hold in every characteristic:

(P1) π is finite of degree 2 and separable, and $K(E)/K(\mathbb{P}^1)$ is Galois with group $\langle\iota\rangle \cong \mathbb{Z}/2$.

(P2) $\iota \neq \mathrm{id}_E$.

(P2) is immediate since E has points of order > 2. For (P1): if $\operatorname{char} K \neq 2$ we may take $E : y^2 = f(x) = x^3 + Ax + B$ with f separable and $\pi = x$, and $\iota(x, y) = (x, -y)$; if $\operatorname{char} K = 2$ then $\iota(x, y) = (x, y + a_1x + a_3)$ with $(a_1, a_3) \neq (0, 0)$, which is again a separable involution fixing x. In both cases $[K(E) : K(x)] = 2$ and ι generates the automorphism group of the extension.

Definition 2.1 (Summation polynomials)**.** $S_2(X_1, X_2) = X_1 - X_2$, and for $m \geq 3$, $S_m \in K[X_1, \ldots, X_m]$ is the reduced defining polynomial of the Zariski closure of

$$\big\{(\pi(Q_1), \ldots, \pi(Q_m)) \;:\; Q_i \in E(\overline{K}),\ Q_1 + \cdots + Q_m = O\big\} \subseteq (\mathbb{P}^1)^m.$$

Equivalently ([7]; see also [6, Prop. 2.1]), for $\xi_1, \ldots, \xi_m \in \overline{K}$ one has $S_m(\xi_1, \ldots, \xi_m) = 0$ if and only if there are $Q_i \in E(\overline{K})$ with $\pi(Q_i) = \xi_i$ and $\sum_i Q_i = O$. For $\operatorname{char} K \neq 2, 3$ and $E : y^2 = x^3 + Ax + B$,

$$\begin{aligned} S_3(X_1, X_2, X_3) = (X_1 - X_2)^2 X_3^2 - 2\big((X_1 + X_2)(X_1X_2 + A) + 2B\big)X_3 \\ + \big((X_1X_2 - A)^2 - 4B(X_1 + X_2)\big), \end{aligned} \tag{1}$$

and S_m for $m \geq 4$ is obtained from the resultant recursion

$$S_m(X_1, \ldots, X_m) = \operatorname{Res}_X\big(S_{m-1}(X_1, \ldots, X_{m-2}, X),\ S_3(X_{m-1}, X_m, X)\big). \tag{2}$$

S_m is symmetric and absolutely irreducible, of degree 2^{m-2} in each variable for $m \geq 3$ [7, 5].

We now name the cover. Fix $m \geq 3$ and put $n := m - 1 \geq 2$.

Definition 2.2 (The summation cover)**.** Let $t_1, \dots, t_n$ be independent transcendentals over K and set

$$L := K(t_1, \dots, t_n) = K\big((\mathbb{P}^1)^n\big), \qquad M := K(E^n).$$

Let $\Gamma := \langle \iota \rangle^n \cong (\mathbb{Z}/2)^n$ act on E^n coordinatewise, let $\Delta := \langle (\iota, \dots, \iota) \rangle \cong \mathbb{Z}/2$ be its diagonal subgroup, and write $\sigma_i \in \Gamma$ for the generator acting by ι in the i-th coordinate only. Let $s\colon E^n \to E$, $(Q_1, \dots, Q_n) \mapsto Q_1 + \cdots + Q_n$, be the summation morphism and put $\varphi := \pi \circ s\colon E^n \to \mathbb{P}^1$. Finally set

$$X_n := E^n/\Delta, \qquad \rho\colon X_n \to (\mathbb{P}^1)^n$$

the map induced by π^n (which is Γ-invariant, hence Δ-invariant).

We identify L with $K(t_1, \dots, t_n)$ via $t_i = \pi \circ \mathrm{pr}_i$, and write $P_i \in E(M)$ for the i-th tautological point, i.e. the i-th projection of the generic point of E^n; thus $\pi(P_i) = t_i$ and $\{\pm P_i\}$ is exactly the fibre of π over t_i.

Lemma 2.3. *M/L is Galois with* $\mathrm{Gal}(M/L) = \Gamma \cong (\mathbb{Z}/2)^n$.

Proof. Γ acts faithfully on E^n by (P2): if $(\varepsilon_1, \dots, \varepsilon_n)$ acts trivially then each ε_i acts trivially on E, so $\varepsilon_i = \mathrm{id}$. Hence by Artin's theorem M/M^Γ is Galois with group Γ. Finally $M^\Gamma = K(E^n/\Gamma) = K\big((E/\langle\iota\rangle)^n\big) = K((\mathbb{P}^1)^n) = L$, using $E^n/\Gamma = (E/\langle\iota\rangle)^n$ and (P1). □

Note what Lemma 2.3 replaces. In coordinates one would argue that $f(t_1), \dots, f(t_n)$ are independent in $L^\times/(L^\times)^2$; that argument is correct for $\mathrm{char}\, K \neq 2$ but needs the Weierstrass model and a squarefreeness check. The fibre-product formulation gives the same conclusion with no computation and in every characteristic.

For $\varepsilon \in \{\pm 1\}^n$ write

$$\theta_\varepsilon \;:=\; \pi\big(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n\big) \;\in\; \mathbb{P}^1(M), \tag{3}$$

so that $\theta_{-\varepsilon} = \theta_\varepsilon$, and $\sigma \in \Gamma$ acts by $\sigma(\theta_\varepsilon) = \theta_{\sigma\varepsilon}$ where $\Gamma = \{\pm 1\}^n$ acts on $\{\pm 1\}^n$ by coordinatewise multiplication.

3 The structure theorem

3.1 The two lemmas

Lemma 3.1 (Root identification)**.** *Let $\Omega \supseteq M$ be an algebraically closed field and $\theta \in \Omega$. Then $S_m(t_1, \dots, t_n, \theta) = 0$ if and only if $\theta = \theta_\varepsilon$ for some $\varepsilon \in \{\pm 1\}^n$. Moreover $\theta_\varepsilon = \infty$ if and only if $\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n = O$.*

Proof. By Definition 2.1 applied over Ω, $S_m(t_1, \dots, t_n, \theta) = 0$ iff there exist $Q_1, \dots, Q_n, R \in E(\Omega)$ with $\pi(Q_i) = t_i$, $\pi(R) = \theta$ and $Q_1 + \cdots + Q_n + R = O$. The fibre of π over t_i is $\{\pm P_i\}$, so $Q_i = \varepsilon_i P_i$ for some sign, and then $R = -\sum_i \varepsilon_i P_i$, whence $\theta = \pi(R) = \theta_\varepsilon$. The converse is the same computation read backwards. The last sentence is $\pi^{-1}(\infty) = \{O\}$. □

Lemma 3.2 (Generic separability)**.** *The 2^{n-1} values θ_ε, $\varepsilon \in \{\pm 1\}^n/\{\pm 1\}$, are pairwise distinct, and none of them equals ∞.*

Proof. For $\emptyset \neq S \subseteq \{1,\dots,n\}$ and $\varepsilon \in \{\pm 1\}^S$ let

$$a_{S,\varepsilon}\colon E^n \longrightarrow E, \qquad (Q_1,\dots,Q_n) \longmapsto \sum_{i\in S} \varepsilon_i Q_i.$$

Each $a_{S,\varepsilon}$ is a surjective homomorphism of group varieties: already its restriction to the i-th factor for one $i \in S$, namely $Q \mapsto \varepsilon_i Q$, is surjective. Hence for any closed subscheme $Z \subsetneq E$ of dimension 0 the preimage $a_{S,\varepsilon}^{-1}(Z)$ is closed of dimension $n-1$, so it is a proper closed subscheme of E^n.

Now suppose $\theta_\varepsilon = \theta_{\varepsilon'}$ with $\varepsilon' \not\equiv \pm\varepsilon$. By (3) this means $\sum_i \varepsilon_i P_i = \pm \sum_i \varepsilon'_i P_i$. In the $+$ case, subtracting gives $\sum_{i\in S} 2\varepsilon_i P_i = O$ with $S = \{i : \varepsilon_i \neq \varepsilon'_i\}$, i.e. $\sum_{i\in S} \varepsilon_i P_i \in E[2]$; in the $-$ case, adding gives the same conclusion with S replaced by its complement. Since $\varepsilon' \neq \varepsilon$ and $\varepsilon' \neq -\varepsilon$, both S and its complement are nonempty. Likewise $\theta_\varepsilon = \infty$ means $a_{\{1,\dots,n\},\varepsilon}$ sends the generic point to O.

So each failure would place the generic point of E^n in one of the finitely many proper closed subschemes $a_{S,\varepsilon}^{-1}(E[2])$ (for $\emptyset \neq S \subsetneq \{1,\dots,n\}$) or $a_{\{1,\dots,n\},\varepsilon}^{-1}(O)$. As E^n is integral, its generic point lies in no proper closed subscheme. □

Remark 3.3. $E[2]$ is a finite group scheme of order 4 in every characteristic — in characteristic 2 it is non-reduced — and Lemma 3.2 only uses that it is finite, so no characteristic is excluded.

3.2 The theorem

Theorem 3.4. *Let K be any field, E/K an elliptic curve, $m \geq 3$ and $n = m-1$. Then, with the notation of §2:*

1. *$S_m(t_1,\dots,t_n,T)$ is separable of degree 2^{m-2} over $L = K(t_1,\dots,t_n)$, with root set $\{\theta_\varepsilon\}$.*
2. *Its splitting field over L is $N = M^\Delta = K(X_n)$, the function field of E^n/Δ.*
3. *N/L is Galois with*

$$\mathrm{Gal}\big(S_m(t_1,\dots,t_n,T)/L\big) \;=\; \Gamma/\Delta \;\cong\; (\mathbb{Z}/2)^{m-2},$$

   acting simply transitively on the 2^{m-2} roots. In particular the cover $\rho\colon X_n \to (\mathbb{P}^1)^n$ is Galois.
4. *K is algebraically closed in N; consequently the arithmetic and geometric monodromy groups agree,*

$$\mathrm{Mon}^{\mathrm{arith}} \;=\; \mathrm{Mon}^{\mathrm{geom}} \;\cong\; (\mathbb{Z}/2)^{m-2}.$$

All four statements are independent of E, of K and of $\operatorname{char} K$.

Proof. (1) By Lemma 3.1 the roots of $S_m(t_1,\dots,t_n,T)$ in an algebraic closure of M are among the θ_ε, of which there are at most $2^{n-1} = 2^{m-2}$ distinct ones; by Lemma 3.2 there are exactly that many, all finite. Since $\deg_T S_m = 2^{m-2}$ (Definition 2.1), these account for all roots with multiplicity one, so $S_m(t_1,\dots,t_n,T)$ is separable and its roots are exactly the θ_ε.

(2), (3) By Lemma 2.3, $\mathrm{Gal}(M/L) = \Gamma$, and by (3) Γ permutes the roots by $\sigma\cdot\theta_\varepsilon = \theta_{\sigma\varepsilon}$. The kernel of this permutation action is

$$\{\sigma \in \{\pm1\}^n : \theta_{\sigma\varepsilon} = \theta_\varepsilon \text{ for all } \varepsilon\} = \{\sigma : \sigma\varepsilon \equiv \pm\varepsilon \text{ for all } \varepsilon\} = \Delta,$$

where the middle equality uses Lemma 3.2 (distinctness) and the last one is immediate: $\sigma\varepsilon = \varepsilon$ for all ε forces $\sigma = (1,\dots,1)$, and $\sigma\varepsilon = -\varepsilon$ for all ε forces $\sigma = (-1,\dots,-1)$. Hence

Γ acts on the 2^{m-2} roots through the faithful quotient Γ/Δ, which has order 2^{m-2} and is therefore *regular* (simply transitive) on them.

Let $\theta := \theta_{(1,\dots,1)}$. Its stabilizer in Γ is Δ, so $[L(\theta) : L] = [\Gamma : \Delta] = 2^{m-2}$ and $L(\theta) = M^{\Delta}$. Since Γ is abelian, $\Delta \trianglelefteq \Gamma$, so M^{Δ} is Galois over L; therefore M^{Δ} already contains all conjugates of θ, i.e. all roots, and it is generated by θ. Thus $N = L(\theta) = M^{\Delta} = K(E^n/\Delta) = K(X_n)$ and $\mathrm{Gal}(N/L) = \Gamma/\Delta \cong (\mathbb{Z}/2)^{m-2}$.

(4) E is geometrically integral, hence so is E^n, hence so is its quotient $X_n = E^n/\Delta$ by the finite group Δ. A geometrically integral K-variety has K algebraically closed in its function field, so $K \cap N = K$ and the constant field extension of N/L is trivial. Therefore $\mathrm{Mon}^{\mathrm{geom}} = \mathrm{Gal}(N\overline{K}/L\overline{K}) = \mathrm{Gal}(N/L) = \mathrm{Mon}^{\mathrm{arith}}$. □

Remark 3.5 (Why the group cannot depend on the curve)**.** Statement (4) is the operative one for cryptanalysis. Arithmetic monodromy strictly larger than geometric monodromy is exactly the situation in which the splitting behaviour of a cover can depend on the base field — and hence, over a finite field, on the curve. Theorem 3.4(4) says that situation does not arise here: the entire cover is defined by the deck group of $E^n \to (E/\pm)^n$, which is $(\mathbb{Z}/2)^n$ for structural reasons that see nothing about E beyond the fact that $\iota \neq \mathrm{id}$.

3.3 Corollaries

Corollary 3.6 (Irreducibility)**.** *$S_m(t_1, \dots, t_n, T)$ is irreducible over L and over $\overline{K}(t_1, \dots, t_n)$.*

Proof. Transitivity of the Galois action, plus Theorem 3.4(4) for the geometric statement. □

This recovers, rather than improves on, the known absolute irreducibility of S_m [7, 5]; it is recorded because it is the same statement read off the group.

Corollary 3.7 (Imprimitivity and block systems)**.** *For $m = 3$ the group is $\mathbb{Z}/2$ acting on 2 points, which is primitive for degree reasons. For $m \geq 4$ the group is abelian and regular of composite degree $2^{m-2} \geq 4$, hence* imprimitive*; its systems of blocks are in bijection with the subgroups $H \leq (\mathbb{Z}/2)^{m-2}$, the blocks being the cosets of H.*

For each $\emptyset \neq I \subsetneq \{1, \dots, n\}$ the map $\theta_\varepsilon \mapsto \pi\big(\sum_{i \in I} \varepsilon_i P_i\big)$ is well defined on roots and its level sets form the block system attached to $H_I = \{\sigma \in \{\pm 1\}^n : \sigma|_I = \pm\mathbf{1}\}/\Delta$. These are the block systems exhibited by the resultant recursion (2)*. For $m \geq 5$ they are not all of them: $(\mathbb{Z}/2)^{m-2}$ has subgroups not of the form H_I.*

The point of Corollary 3.7 is deflationary. Imprimitivity of a solving system is sometimes a lever — it can permit a resolvent of smaller degree — but here the block systems are exactly the sign-subgroup systems already visible in the recursion that defines S_m, and the associated resolvents are summation polynomials of smaller arity. Nothing new is exposed.

Corollary 3.8 (Discriminant square class)**.** *For $m = 3$, $\mathrm{disc}_T\, S_3(t_1, t_2, T) = 16\, f(t_1) f(t_2)$ (in* char $\neq 2, 3$*), which is not a square in L. For every $m \geq 4$, $\mathrm{disc}_T\, S_m(t_1, \dots, t_n, T)$ is a square in L.*

Proof. The $m = 3$ identity is an identity in $\mathbb{Z}[X_1, X_2, A, B]$; it follows from (1) by expansion, or conceptually: the two roots are $x(P_1 + P_2)$ and $x(P_1 - P_2)$, and the chord formula gives $x(P_1 + P_2) - x(P_1 - P_2) = -4y_1y_2/(t_1 - t_2)^2$, whence $\mathrm{disc} = c_2^2(\theta_+ - \theta_-)^2 = 16y_1^2y_2^2 = 16f(t_1)f(t_2)$ with $c_2 = (t_1 - t_2)^2$ the leading coefficient. It is not a square because $f(t_1)f(t_2)$ is squarefree in the unique factorization domain $\overline{K}[t_1, t_2]$ (its six irreducible factors are pairwise distinct precisely because E is nonsingular).

For $m \geq 4$: the discriminant of a separable polynomial is a square in the base field if and only if the Galois group is contained in the alternating group. By Theorem 3.4 the group

acts regularly, so every nonidentity element is a fixed-point-free involution on 2^{m-2} points, i.e. a product of 2^{m-3} transpositions. For $m \geq 4$ we have $2^{m-3} \geq 2$, an even number, so every element is even. For $m = 3$, $2^{m-3} = 1$ is odd, which is the exception. □

Corollary 3.8 is worth stating because it locates the $m = 3$ discriminant identity precisely: it is not the first case of a family but the unique case in which the discriminant carries information. For $m \geq 4$ the quadratic character of $\operatorname{disc}_T S_m$ is constant, and any statistic built on it is vacuous.

3.4 The branch locus

Lemma 3.2 also describes exactly where the cover degenerates. Over a point $(a_1, \ldots, a_n)$ of the base, the fibre fails to consist of 2^{m-2} distinct affine points precisely when one of the closed conditions in the proof holds. Explicitly:

Proposition 3.9. *Let $a = (a_1, \ldots, a_n) \in \overline{K}^n$ and P_i a point with $\pi(P_i) = a_i$. Then*

1. *$\deg_T S_m(a_1, \ldots, a_n, T) < 2^{m-2}$ (a root at infinity) if and only if $S_n(a_1, \ldots, a_n) = 0$, i.e. some signed sum $\sum_i \varepsilon_i P_i$ equals O;*
2. *two roots collide if and only if $\sum_{i \in S} \varepsilon_i P_i \in E[2]$ for some $\emptyset \neq S \subsetneq \{1, \ldots, n\}$ and some signs. The case $S = \{i\}$ is $P_i \in E[2]$, i.e. a_i is a branch point of π.*

The union of these loci is a proper closed subset of $(\mathbb{P}^1)^n$ of codimension 1, so it has $O_m(q^{n-1})$ points over $\mathbb{F}_q$.

For $m = 3$ this is completely explicit: $n = 2$, the only proper nonempty subsets are $\{1\}$ and $\{2\}$, so the degenerate locus is $\{t_1 = t_2\} \cup \{f(t_1)f(t_2) = 0\}$, the first component being $S_2(t_1, t_2) = t_1 - t_2 = 0$. Over $\mathbb{F}_p$ its point count is $p + Z(2p - Z) - Z$ where $Z = \#\{x : f(x) = 0\} \in \{0, 1, 3\}$, which is exactly what is measured in §8.

3.5 Special j-invariants, complex multiplication, and characteristic

Three families are worth checking explicitly, because they are the standard sources of exceptional behaviour and because a claim of the form "for every curve" must survive them.

Extra automorphisms ($j = 0$, $j = 1728$). If $\operatorname{char} K \neq 2, 3$ then $\operatorname{Aut}(E, O) = \mu_2$ except for $j = 1728$ (μ_4) and $j = 0$ (μ_6). Extra automorphisms do not enter Theorem 3.4 at any point: the construction uses only ι and the quotient π, and Lemma 2.3 needs only (P1) and (P2). What an extra automorphism $\zeta \in \operatorname{Aut}(E, O)$ does is act on $E/\langle\iota\rangle = \mathbb{P}^1$, hence on the *base* $(\mathbb{P}^1)^n$, commuting with ρ. It is a symmetry of the cover over an automorphism of the base, not a deck transformation of ρ, so it does not enlarge or shrink the monodromy group. This is verified numerically in §8 on $j = 0$ curves with $p \equiv 1$ and $p \equiv 2 \pmod 3$ and on $j = 1728$ curves with $p \equiv 1$ and $p \equiv 3 \pmod 4$ — i.e. with the extra automorphisms rational and irrational.

Complex multiplication and supersingularity. Likewise irrelevant. $\operatorname{End}(E)$ never appears; the deck group of $E^n \to (E/\pm)^n$ is $(\mathbb{Z}/2)^n$ whether or not E has CM, and whether or not E is supersingular. A supersingular curve is included in the verification battery.

Characteristic. Theorem 3.4 is stated and proved in every characteristic, because it uses only (P1) and (P2). Three caveats are nevertheless real and are stated rather than hidden.

- The *Weierstrass computations* — (1), the explicit $\operatorname{disc}_T S_3 = 16 f(t_1) f(t_2)$, and the description of the branch locus by $f(t_i) = 0$ — require $\operatorname{char} K \neq 2, 3$. In characteristic 2 and 3 the corresponding summation polynomials must be taken in the appropriate model [4]; the group-theoretic conclusion is unchanged but the formulas are not.
- In characteristic 2, $E[2]$ has order 2 (ordinary) or is infinitesimal (supersingular) as a group scheme; Lemma 3.2 is unaffected, but Corollary 3.8 is not meaningful, since "square class of the discriminant" is the wrong invariant in characteristic 2.
- Theorem 4.2 below is stated over $\mathbb{F}_q$ using a splitting invariant $c(\cdot) \in \mathbb{Z}/2$ that is characteristic-free; only its identification with a quadratic character requires q odd.

4 An exact factorization law over a finite field

Let $K = \mathbb{F}_q$ now. For $a \in \mathbb{P}^1(\mathbb{F}_q)$ define the *splitting invariant*

$$c(a) := \begin{cases} 0 & \text{if } \pi^{-1}(a) \subseteq E(\mathbb{F}_q) \text{ and } \#\pi^{-1}(a) = 2, \\ 1 & \text{if } \pi^{-1}(a) \cap E(\mathbb{F}_q) = \emptyset, \\ * & \text{if } \#\pi^{-1}(a) = 1 \quad (a \text{ a branch point}). \end{cases} \tag{4}$$

For q odd and $E : y^2 = f(x)$ one has $c(a) = 0$ if $\chi(f(a)) = +1$, $c(a) = 1$ if $\chi(f(a)) = -1$ and $c(a) = *$ if $f(a) = 0$, where χ is the quadratic character of $\mathbb{F}_q$. Thus $c(a)$ records whether Frobenius fixes or swaps the two points above a.

Definition 4.1. Call $a = (a_1, \dots, a_n) \in \mathbb{F}_q^n$ *good* if $S_n(a_1, \dots, a_n) \neq 0$ and $\sum_{i \in S} \varepsilon_i P_i \notin E[2]$ for every $\emptyset \neq S \subsetneq \{1, \dots, n\}$ and every choice of signs, where $P_i \in E(\mathbb{F}_{q^2})$ is either point above a_i. By Proposition 3.9, a is good if and only if $S_m(a_1, \dots, a_n, T)$ is separable of degree 2^{m-2}; the non-good locus has $O_m(q^{n-1})$ points. Note $c(a_i) \neq *$ for good a.

Theorem 4.2 (Factorization law)**.** *Let $a \in \mathbb{F}_q^n$ be good. Then $S_m(a_1, \dots, a_n, T) \in \mathbb{F}_q[T]$ is squarefree of degree 2^{m-2} and exactly one of the following holds.*

1. *If $c(a_1) = c(a_2) = \cdots = c(a_n)$, then $S_m(a_1, \dots, a_n, T)$ splits into 2^{m-2} distinct linear factors over $\mathbb{F}_q$, with roots $\{x(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n)\}$.*
2. *Otherwise $S_m(a_1, \dots, a_n, T)$ is a product of 2^{m-3} distinct irreducible quadratics over $\mathbb{F}_q$, and has* no *root in $\mathbb{F}_q$.*

In particular: no irreducible factor of degree ≥ 3 ever occurs; all irreducible factors always have the same degree; and the factorization type depends on a only through the class of $(c(a_1), \dots, c(a_n))$ in $(\mathbb{Z}/2)^n/\Delta$.

Proof. Goodness says the specialization lies in the étale locus of the cover $\rho\colon X_n \to (\mathbb{P}^1)^n$, so the factorization type of the fibre equals the cycle type of the Frobenius element in $\operatorname{Gal}(N/L) = \Gamma/\Delta$ acting on the 2^{m-2} roots (Dedekind's theorem for specializations of a Galois cover; see e.g. [3, Ch. 6]).

Frobenius acts on $M = L(E^n)$ through its action on each factor, and on the i-th factor it fixes or swaps the two points of $\pi^{-1}(a_i)$ according to $c(a_i)$: that is, the Frobenius element of $\Gamma = \{\pm 1\}^n$ is $\big((-1)^{c(a_1)}, \dots, (-1)^{c(a_n)}\big)$. Its image in Γ/Δ is trivial exactly when all $c(a_i)$ agree, which is case (1); then Frobenius fixes every root and the fibre is totally split, and the roots are the signed sums by Lemma 3.1.

Otherwise the Frobenius element is a nontrivial element of Γ/Δ, which acts *regularly* on the roots by Theorem 3.4(3). A nontrivial element of a regular permutation group has no fixed point, and here it has order 2, so its cycle type is $2^{2^{m-3}}$: the fibre is a disjoint union of 2^{m-3} Frobenius orbits of size 2, giving 2^{m-3} distinct irreducible quadratic factors and no rational root. □

Corollary 4.3 (Exact counts and density)**.** *Let $S = \#\{a \in \mathbb{F}_q : c(a) = 0\}$, $N = \#\{a : c(a) = 1\}$ and $Z = \#\{a : c(a) = *\}$, so $S + N + Z = q$ and $\#E(\mathbb{F}_q) = 1 + Z + 2S$. For $\varepsilon \in \{\pm 1\}^n$ with k entries equal to $+1$, the number of $a \in \mathbb{F}_q^n$ whose splitting-invariant vector lies in the class of ε is exactly*

$$S^k N^{n-k} + N^k S^{n-k}. \tag{5}$$

Consequently the number of good *a giving a totally split fibre is $S^n + N^n$ minus the number of non-good a with all $c(a_i)$ equal, and the density of totally split fibres satisfies*

$$\Big|\frac{S^n + N^n}{q^n} - 2^{-(m-2)}\Big| \;\le\; \frac{31(m-1)^2}{2^{m-1}\,q} \qquad \text{for } q \ge 9(m-1)^2.$$

Proof. (5) is immediate. For the density write $S/q = (1-\alpha)/2$ and $N/q = (1+\beta)/2$ with $\alpha = (t+Z)/q$, $\beta = (t-Z)/q$ and $t = q+1-\#E(\mathbb{F}_q)$ the trace, so $|\alpha|, |\beta| \le (2\sqrt{q}+3)/q =: \delta$ by Hasse. Then

$$\frac{S^n + N^n}{q^n} = 2^{-n}\big[(1-\alpha)^n + (1+\beta)^n\big] = 2^{-n}\Big[2 + n(\beta-\alpha) + \textstyle\sum_{k\ge 2}\binom{n}{k}\big((-\alpha)^k + \beta^k\big)\Big].$$

The linear term is $n(\beta-\alpha) = -2nZ/q$, which is *independent of the trace*: the Hasse-interval contribution cancels at first order. The tail is bounded by $2\big[(1+\delta)^n - 1 - n\delta\big] \le (n\delta)^2 e^{n\delta}$, and $Z \le 3$, giving $\big|(S^n+N^n)/q^n - 2^{1-n}\big| \le 2^{-n}\big[6n/q + 9n^2 e^{3n/\sqrt{q}}/q\big]$, which for $q \ge 9n^2$ is at most $2^{-n}\cdot 31n^2/q$. □

Remark 4.4. The cancellation of the trace term is the general form of the $m = 3$ observation that the split frequency is $\frac{1}{2} + O(1/q)$ rather than $\frac{1}{2} + O(1/\sqrt{q})$: the two error terms $\pm t/q$ enter S^n and N^n with opposite signs and cancel to first order at every n. So the split density carries no first-order information about the curve — a further sense in which this statistic is not a lever.

5 Specialization to the factor-base locus

Index-calculus relation search over $\mathbb{F}_p$ works with a factor base $\mathcal{F} \subseteq E(\mathbb{F}_p)$ and looks for relations $P_1 + \cdots + P_{m-1} = R$ with $P_i \in \mathcal{F}$. The coordinates that enter the summation polynomial are therefore x-coordinates of *rational* points — a proper subvariety's worth of the parameter space, of density about $1/2$ in each coordinate.

Theorem 5.1 (Complete splitting on the factor-base locus)**.** *Let $a_1, \ldots, a_n \in \mathbb{F}_q$ be such that each $a_i = x(P_i)$ for some $P_i \in E(\mathbb{F}_q)$, and suppose a is good. Then $S_m(a_1, \ldots, a_n, T)$ splits into 2^{m-2} distinct linear factors over $\mathbb{F}_q$, with root set exactly*

$$\big\{\, x(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n) \;:\; \varepsilon \in \{\pm 1\}^n \,\big\} \;\subseteq\; \mathbb{F}_q,$$

at every $m \ge 3$. Equivalently, the decomposition group of the cover ρ at every such point is trivial: the factor-base locus is contained in the totally split locus.

Proof. $c(a_i) = 0$ for all i, so Theorem 4.2(1) applies. Directly: the P_i are $\mathbb{F}_q$-rational, hence so is every signed sum, hence so is every root. □

5.1 Why complete splitting is not an algorithm

A reader meeting Theorem 5.1 for the first time may suspect it trivializes relation search. It does not, and the reason is an asymmetry between which variables are given and which are sought.

Theorem 5.1 is a statement about solving for the *last* coordinate given the first $m-1$. Its operational content is: choose $P_1, \dots, P_{m-1} \in \mathcal{F}$; then the 2^{m-2} admissible values of the remaining coordinate are $x(\pm P_1 \pm \cdots \pm P_{m-1})$, computable by $O(2^{m-2})$ group operations. That is precisely the "solve for the last variable and test membership" step that index calculus already performs, and it is why the summation-polynomial system is usually posed in $m-1$ rather than m unknowns.

Relation search asks the opposite question. The target $x(R)$ is *fixed*, and one must find $a_1, \dots, a_{m-1}$ *in the factor base* with $S_m(a_1, \dots, a_{m-1}, x(R)) = 0$. Complete splitting in the last variable says nothing about the existence or the cost of finding such a tuple: it does not make the factor base membership test cheaper, it does not change how many tuples must be tried, and it does not change the algebraic difficulty of the system when one does pose it as a polynomial system. The cost of index calculus sits in that search and in the subsequent linear algebra, neither of which Theorem 5.1 touches.

What Theorem 5.1 does give is negative and methodological, and it is the reason the theorem is stated at all:

Corollary 5.2 (The census obstruction)**.** *Any statistic of the form "the factorization type, cycle type, or Frobenius class of the fibre of S_m" is* identically constant *on the factor-base locus, at every m and for every curve. In particular, a Chebotarev-style census of generic fibres — sampling $(a_1, \dots, a_{m-1})$ uniformly from $\mathbb{F}_q^{m-1}$ — measures a quantity whose value on the locus where relation search operates is known in advance and is the same for all curves.*

This closes a natural but unproductive line of enquiry: one cannot find a curve family with anomalous relation rates by looking for anomalous monodromy or anomalous fibre statistics, because by Theorem 3.4 there is no anomalous monodromy to find, and by Theorem 5.1 the fibre statistic is constant where it would have to matter.

6 Conservation of decomposition yield

The second half of the synthesis is combinatorial and even more elementary. It is recorded here because it constrains the same design space from the other side.

Proposition 6.1 (Conservation)**.** *Let G be a finite abelian group of order N and let $\mathcal{D} \subseteq G \setminus \{0\}$ have B distinct elements. For $r \in G$ let $c_{\mathcal{D}}(r)$ be the number of size-m multisets from $\mathcal{D}$ summing to r. Then*

$$\sum_{r \in G} c_{\mathcal{D}}(r) = \binom{B+m-1}{m}, \qquad \textit{hence} \qquad \mathbb{E}_r\big[c_{\mathcal{D}}(r)\big] = \frac{1}{N}\binom{B+m-1}{m}$$

for every *$\mathcal{D}$ of size B, independently of how $\mathcal{D}$ is chosen.*

Proof. Every size-m multiset from $\mathcal{D}$ sums to exactly one element of G; count the pairs (multiset, its sum) in two ways. □

Corollary 6.2. *At matched factor-base size B:*

1. *The mean decomposition yield per target is a fixed function of B, m and N. Any hypothesis asserting that some factor-base geometry has a mean-yield advantage — or that such an advantage grows with N — is refuted by arithmetic, with no experiment.*

2. *Only the* distribution *of $c_{\mathcal{D}}$ is free. Coverage, the fraction of targets admitting at least one decomposition, satisfies* $\mathrm{cov} \le \min(1, \mu)$ *with μ the mean, with equality iff no target has two decompositions. Additive structure that creates repeated sums therefore* lowers *coverage at matched size.*

3. *Typing the decomposition (requiring the i-th summand to come from a sub-base $\mathcal{D}_i$ with $\sum_i B_i = B$) is a fixed penalty, not a lever: $\prod_i B_i \le (B/m)^m$, strictly below the untyped $\binom{B+m-1}{m} \approx B^m/m!$.*

Proposition 6.1 is a one-line double count and no novelty is claimed for it. It is included because, together with §5, it delimits the design space from both sides, and because it is easy to state hypotheses that it refutes without noticing.

Remark 6.3 (What conservation does not say)**.** It does not say structured bases cannot beat random ones: the coverage headroom up to $\min(1, \mu)$ is real and attainable (a Sidon/B_h base attains it). It says nothing about the *cost of finding* a decomposition or about the linear algebra — which is where the cost of index calculus actually sits. And it is entirely consistent with index calculus working: the Gaudry–Diem $1/n!$ decomposition probability over extension fields *is* this conservation mean; the extension-field advantage lives in the solve, not in the yield.

7 Synthesis: which levers are fixed and which are free

Quantity	**Status**
Mean m-term decomposition yield at fixed factor-base size	**Fixed** by conservation (Prop. 6.1); identical for all factor bases of the same size.
Galois group / monodromy of the summation cover	**Fixed** = $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic (Thm. 3.4); arithmetic = geometric.
Fibre factorization type on the factor-base locus	**Fixed** = totally split, at every m (Thm. 5.1).
Fibre factorization type at a uniform random point	**Fixed** up to $O(m^2/q)$: totally split with density $2^{-(m-2)}$, else 2^{m-3} quadratics (Thm. 4.2, Cor. 4.3); trace cancels at first order.
Square class of $\mathrm{disc}_T\, S_m$	**Fixed** = square for $m \ge 4$; informative only at $m = 3$ (Cor. 3.8).
Block systems of the fibre	**Fixed** = subgroups of $(\mathbb{Z}/2)^{m-2}$; the recursion already exhibits the sign-subgroup family (Cor. 3.7).
Coverage, and the multiplicity distribution	**Free**, bounded by $\min(1, \mu)$; attainable headroom is a constant factor, not an exponent.
Relation rank and independence	**Free**; invisible to all of the above (the mean scores a rank-deficient base as tied with a random one).
Recognizability: cost of testing factor-base membership	**Free**.
Solve cost: the algebraic cost of the decomposition system	**Free**, and this is where the cost of index calculus sits.

The headline is the separation. *Generic-fibre arithmetic statistics and factor-base relation statistics are different objects.* The first are governed by Theorem 3.4 and are the same for every curve; the second live on a locus where, by Theorem 5.1, the first are constant. A proposal that hopes to raise the relation rate by changing curve or factor base must argue in the "free" half of the table; the "fixed" half is closed by the results above, and a proposal arguing there can be refuted on sight.

Remark 7.1 (Scope of "moves no exponent")**.** The per-fibre statements above are at fixed m and fixed field. A factor of 2 applied once per summation fibre in a Gaudry/Diem-style decomposition over $\mathbb{F}_{q^n}$ with n growing compounds as 2^{n-1} and *would* move an exponent. The claims of this section are made at fixed m over a prime field and must not be quoted without that scope.

8 Computational verification

Every statement above that can be checked at finite scale was checked, on a battery of 20 curves over the primes $101, 103, 211, 307, 401, 1009$: twelve of generic j; one $j = 0$ with $p \equiv 1 \bmod 3$ and one with $p \equiv 2 \bmod 3$; one $j = 1728$ with $p \equiv 1 \bmod 4$ and one supersingular with $p \equiv 3 \bmod 4$; and four with full rational 2-torsion. The implementation is self-contained (no computer-algebra system): summation polynomials are built from (1) and (2) by Sylvester determinants recovered from $2^{m-2}+1$ evaluations, signed sums are computed by elliptic-curve arithmetic over $\mathbb{F}_{p^2}$, and factorization types are read off by distinct-degree factorization. Seed 20260807; single run, 68 seconds.

The $m = 3$ discriminant identity (Cor. 3.8). Verified as an identity in $\mathbb{Z}[X_1, X_2, A, B]$ with exact integer coefficients — not as a congruence modulo a prime.

Root identification and the factorization law (Thms. 3.4, 4.2). 18 000 uniform specializations at $m = 3, 4, 5$, of which 16 737 were good. On all 16 737: the 2^{m-2} signed sums were distinct, none at infinity, and each was a root of the computed polynomial (16 737/16 737); the factorization type matched the prediction of Theorem 4.2 (16 737/16 737); no irreducible factor of degree > 2 ever appeared; and no fibre ever had a *mixed* factorization type. All 2^{m-2} Frobenius classes were realized at each m, and each class produced exactly one factorization type — $1^{2^{m-2}}$ for the trivial class, $2^{2^{m-3}}$ for every other. The discriminant square class was $+1$ on all 7324 good samples at $m = 4$ and all 1627 at $m = 5$, and mixed at $m = 3$ (3828 square, 3958 non-square), as Corollary 3.8 requires.

Factor-base locus (Thm. 5.1). 18 000 specializations with every a_i the x-coordinate of a rational point, of which 16 107 were good. All 16 107 split completely, with all 2^{m-2} roots in $\mathbb{F}_p$ and equal to the signed sums. No counterexample at any m on any curve.

Exhaustive checks and the exact class count (Cor. 4.3). All p^{m-1} specializations were enumerated for $(m, p) \in \{3\} \times \{53, 101, 211\}$ and $\{4\} \times \{37, 53\}$: 257 061 specializations, 224 906 of them good. Zero factorization-type mismatches. For *every* Frobenius class in every cell, the identity

$$\underbrace{S^k N^{n-k} + N^k S^{n-k}}_{\text{closed form (5)}} = \#\{\text{good } a \text{ in the class}\} + \#\{\text{degenerate } a \text{ in the class}\}$$

held with residual exactly 0. The degenerate locus matched Proposition 3.9 exactly: for $m = 3$ with $Z = 0$ it had exactly p points (101 at $p = 101$, 211 at $p = 211$ — the diagonal $t_1 = t_2$), and for $p = 53$ with $Z = 1$ exactly $157 = 53 + (2 \cdot 53 - 1) - 1$ points.

Null controls. To show the instrument is discriminating rather than vacuous, the same pipeline was run on two degree-4 families that are not summation polynomials. NULL-A, a uniformly random degree-4 polynomial over $\mathbb{F}_{401}$: of 599 separable samples, only 13.7% obeyed the dichotomy of Theorem 4.2, with types $1^1 3^1$, 4^1 and $1^2 2^1$ all common. NULL-B, the S_4 construction (2) with the inner summation polynomial replaced by a random quadratic — the elliptic structure removed and everything else held fixed: of 595 separable samples, 48.9% obeyed it, with 4^1 and $1^2 2^1$ both frequent. The factorization law was checked directly on $16\,737 + 224\,906 = 241\,643$ good specializations of genuine summation polynomials and held on every one; a law obeyed by 100% of those and by 13.7% and 48.9% of matched controls is not an artifact of the measurement.

9 Relation to prior work, and the limits of this novelty assessment

Classical, and claimed as such. Definition 2.1 and the resultant recursion (2) are Semaev's [7]. Lemma 3.1 — the roots of $S_m(a_1, \ldots, a_{m-1}, T)$ are x-coordinates of signed sums — is an immediate consequence of the defining property and is used routinely [5, 2, 6]; no novelty is claimed for it. The degree 2^{m-2}, the symmetry, and the absolute irreducibility of S_m are standard [7, 5], so Corollary 3.6 is a re-derivation, not a new result. The operational fact behind §5.1 — that the last coordinate is determined by group arithmetic — is likewise standard practice. Proposition 6.1 is a double count.

What is offered as possibly new. Theorem 3.4 (the Galois group for all m, with the identification of the splitting field as $K(E^{m-1}/\Delta)$ and the coincidence of arithmetic and geometric monodromy); Theorem 4.2 (the exact factorization dichotomy over $\mathbb{F}_q$ and its dependence only on the class of the splitting-invariant vector); Corollary 3.8 (the discriminant square class, isolating $m = 3$); and the synthesis of §7 with Corollary 5.2.

The limits of this assessment — stated plainly. Prime-field Semaev index calculus has a substantial literature, and the statements above are the kind that could plausibly be folklore among specialists in summation polynomials or in function-field arithmetic. The following is the exact extent of the prior-art search behind this paper, and it is not sufficient to establish novelty:

- Targeted searches on the combination {summation polynomial, Semaev} with {Galois group, monodromy, splitting field, factorization pattern, Kummer quotient, deck group, elementary abelian} returned no work stating either Theorem 3.4 or Theorem 4.2.
- **eprint.iacr.org was not reachable** from the environment in which this work was carried out (HTTP 403 to both the automated fetcher and a direct request). Since that is the primary venue for this literature — including Semaev's original preprint — a substantial fraction of the relevant corpus was *not* inspected. arXiv was reachable and was inspected.
- Consequently novelty is **not adjudicated** here in either direction. Before submission, a reader with access must check at minimum: [7], [5], [1], [2], [4], [6], and the literature on point compression and Kummer-line arithmetic, where the quotient $E^n/\pm$ appears for unrelated reasons and where the deck-group observation may already be recorded.

Outstanding requirement before submission. Independent review of §3 by a reader experienced in function fields or arithmetic geometry has *not* been obtained. The proofs are short and the computational evidence is strong, but the load-bearing steps — Lemma 2.3's use of Artin's theorem for a quotient by a finite group acting faithfully, and Lemma 3.2's properness argument in characteristic 2 where $E[2]$ is non-reduced — deserve a second pair of eyes before this is posted.

10 Open questions

Question 10.1. The block systems of Corollary 3.7 correspond to all subgroups of $(\mathbb{Z}/2)^{m-2}$, while the resultant recursion exhibits only the family H_I. For $m \geq 5$ the others are not visible in the recursion. Do they correspond to resolvents with any computational meaning, or are they only group-theoretic shadows?

Question 10.2. Theorem 3.4 concerns the fibre in one variable with the others generic. Relation search fixes the target and constrains all remaining variables to a factor base. Is there a statistic on *that* configuration — the intersection of $V(S_m)$ with a product of factor-base subschemes — which is not constant across curves? Corollary 5.2 says the fibre statistics are not; it does not say no statistic is.

Question 10.3. Corollary 4.3 shows the trace cancels at first order in the split density at every m. Is there a natural weighting of fibres under which the trace survives?

References

[1] C. Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica* 147(1):75–104, 2011.

[2] J.-C. Faugère, P. Gaudry, L. Huot, G. Renault. Using symmetries in the index calculus for elliptic curves discrete logarithm. *Journal of Cryptology* 27(4):595–635, 2014.

[3] M. D. Fried, M. Jarden. *Field Arithmetic*, 3rd ed. Springer, 2008.

[4] S. D. Galbraith, S. W. Gebregiyorgis. Summation polynomial algorithms for elliptic curves in characteristic two. In *INDOCRYPT 2014*, LNCS 8885, pp. 409–427, 2014.

[5] P. Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation* 44(12):1690–1702, 2009.

[6] M. Kosters, S. L. Yeo. Notes on summation polynomials. arXiv:1503.08001, 2015.

[7] I. Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. IACR Cryptology ePrint Archive, Report 2004/031, 2004.

[8] J. H. Silverman. *The Arithmetic of Elliptic Curves*, 2nd ed. GTM 106, Springer, 2009.